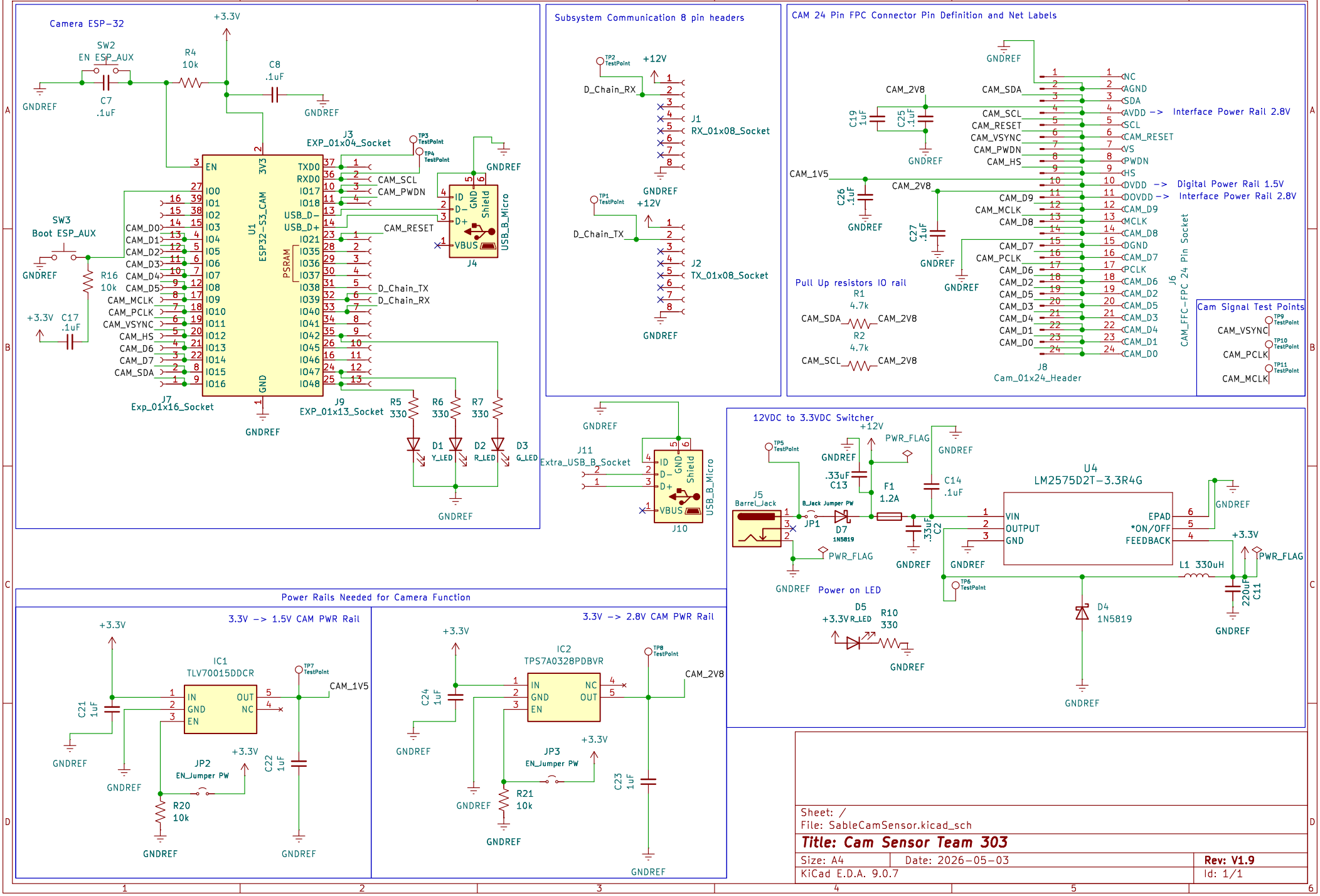

Camera ESP-32
SW2
EN ESP_AUX
R4
10k
C8
.1uF
C7
.1uF
+3.3V
GNDREF
J3
EXP_01x04_Socket
TP3 TestPoint
TP4 TestPoint
CAM_SCL
CAM_PWDN
CAM_RESET
D_Chain_TX
D_Chain_RX
USB_B_Micro
J4
SW3
Boot ESP_AUX
R16
10k
C17
.1uF
U1
ESP32-S3_CAM
PSRAM
CAM_D0
CAM_D1
CAM_D2
CAM_D3
CAM_D4
CAM_D5
CAM_MCLK
CAM_PCLK
CAM_VSYNC
CAM_HS
CAM_D6
CAM_D7
CAM_SDA
J7
Exp_01x16_Socket
J9
EXP_01x13_Socket
R5
330
R6
330
R7
330
D1
Y_LED
D2
R_LED
D3
G_LED
Subsystem Communication 8 pin headers
TP2 TestPoint
+12V
D_Chain_RX
J1
RX_01x08_Socket
TP1 TestPoint
D_Chain_TX
J2
TX_01x08_Socket
CAM 24 Pin FPC Connector Pin Definition and Net Labels
CAM_2V8
C19
1uF
C25
.1uF
CAM_1V5
C26
.1uF
C27
.1uF
Pull Up resistors IO rail
R1
4.7k
R2
4.7k
J8
Cam_01x24_Header
J6
CAM_FFC-FPC 24 Pin Socket
NC
AGND
SDA
AVDD -> Interface Power Rail 2.8V
SCL
CAM_RESET
VS
PWDN
HS
DVDD -> Digital Power Rail 1.5V
DOVDD -> Interface Power Rail 2.8V
MCLK
DGND
PCLK
Cam Signal Test Points
TP9 TestPoint
TP10 TestPoint
TP11 TestPoint
J11
Extra_USB_B_Socket
J10
12VDC to 3.3VDC Switcher
TP5 TestPoint
J5
Barrel_Jack
JP1
B_Jack Jumper PW
D7
1N5819
F1
1.2A
.33uF
C13
C14
.1uF
C2
.33uF
U4
LM2575D2T-3.3R4G
VIN
OUTPUT
GND
EPAD
*ON/OFF
FEEDBACK
L1 330uH
220uF
C11
PWR_FLAG
TP6 TestPoint
Power on LED
D5
R_LED
R10
330
D4
1N5819
Power Rails Needed for Camera Function
3.3V -> 1.5V CAM PWR Rail
IC1
TLV70015DDCR
IN
GND
EN
OUT
NC
C21
1uF
C22
1uF
JP2
EN_Jumper PW
R20
10k
TP7 TestPoint
CAM_1V5
3.3V -> 2.8V CAM PWR Rail
IC2
TPS7A0328PDBVR
C24
1uF
C23
1uF
JP3
EN_Jumper PW
R21
10k
TP8 TestPoint
CAM_2V8
Sheet: /
File: SableCamSensor.kicad_sch
Title: Cam Sensor Team 303
Size: A4
Date: 2026-05-03
Rev: V1.9
KiCad E.D.A. 9.0.7
Id: 1/1